



Module 12 Day 1: 07 Configuring UFW

Activity Introduction

In this activity, you will assume the role of SOC analyst for a large retailer, Better Buys, Inc.

- Since ecommerce is a large part of your organization's business infrastructure, Better Buys takes various forms of online payment, including credit cards. As such, it must comply with the Payment Card Industry Data Security Standard (PCI DSS).
- The PCI DSS has 12 requirements with which organizations must comply. The first states that organizations must protect their systems with firewalls.
- In this activity, you'll configure an uncomplicated firewall (UFW) to ensure that your organization is in compliance with PCI DSS.
- In addition to concerns about being in compliance, one of your junior analysts has detected numerous port scans that originated from public sources. Your organization requires the use of ports 110, 143, 587, 80, and 443 to be open at all times as part of its daily operations.
- To mitigate adversarial reconnaissance and network fingerprinting as well as being PCI DSS compliant, you've decided to harden your network by filtering all non-essential ports on your perimeter firewall.
- Feel free to use online resources and manual pages to help you through these exercises.
 - Use `ufw version` to determine what version of UFW is installed.
 - Use `sudo ufw reset` to reset all UFW rules to factory defaults.
 - Use `sudo ufw status` to check the current status of the firewall.
 - Use `sudo ufw enable` to start the firewall and update rules.
 - Use `sudo ufw default deny incoming` to block all incoming connections.
 - Use `sudo ufw default allow outgoing` to allow all outgoing connections.
 - Use `sudo ufw allow` to open specific ports.

- Use `sudo ufw deny` to close specific ports.
- Use `sudo ufw delete` to delete rules.
- Use `sudo ufw disable` to shut down the firewall.
- Use `sudo ufw reload` to reload the UFW firewall.
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Setup

Complete the following steps before beginning the activity:

1. From your Netsec VM, log into the `ufw` container using `docker exec -it ufw bash`
 - **IMPORTANT:** After you login to the container, you must run the following:

```
sudo sed -i 's/^IPV6=.* /IPV6=no/' /etc/default/ufw
```

2. Reset all the UFW rules to their factory defaults. This will allow us to customize UFW with our own rule sets.
 - Run `sudo ufw reset`.
 - When prompted, enter the letter `y` and press `Enter`.
3. Make sure `firewalld` is not running.
 - Run `sudo service firewalld stop`.
 - Run `sudo service firewalld status`.

Activity Instructions

1. Test that you can SSH into the Ubuntu UFW using Ubuntu `firewalld` (Other container). The `firewalld` container will be used to test connections to the UFW container.
 - Open a new terminal window.
 - Log into the `firewalld` container using an interactive bash session to enter the container use `sudo docker exec -it firewalld bash`.
 - Attempt to use SSH to log into the UFW container.
 - See if you can use SSH to gain access to your Ubuntu UFW. Make sure to use the IP address for the `Eth0` interface of the Ubuntu UFW.
 - Type the command to SSH.
 - username: `sysadmin`
 - password: `cybersecurity`
 - Type `exit` to terminate the connection.

- Now that you know you can log in using SSH, you'll use UFW to stop that from happening.
- 2. Switch back to the Ubuntu UFW. Enable UFW and set up default rules that block all incoming traffic and allow all outgoing traffic.
- 3. Block all incoming SSH connections.
- 4. The protocol Telnet sends unencrypted traffic and is therefore a vulnerability. Configure the firewall to block Telnet.
- 5. Open the following ports to allow incoming information from web servers and mail applications.
 - 80: For HTTP connections
 - 143: For IMAP (Internet Message Access Protocol)
 - 587: For SMTP (Simple Mail Transfer Protocol)
 - 110: For POP (Post Office Protocol)
 - 443: For HTTPS (HTTP over SSL)
- 6. Use the SSH protocol to try to connect to the UFW container from firewalld container.
 - Type the SSH command to connect to the UFW machine.
 - Press Ctrl+C to stop your terminal from processing.
 - Does this confirm that the Ubuntu UFW container will block all incoming SSH traffic?
- 7. Recall that you blocked Telnet but opened up port 80. What will happen if you try to use Telnet on port 80?
 - Type the command to telnet to port 80.
 - What do you see? What happens if you type a letter and then hit enter?
 - You will see the web server contacted on port 80 and the HTTP headers retrieved.
 - Why is it able to make the connection with Telnet even though you blocked Telnet?
- 8. Switch back to the UFW container. Delete your initial rule blocking SSH. It is no longer needed because you are actively blocking all incoming connections.
 - Type the command that deletes the first rule only: `sudo ufw delete #`. (It goes by the line the rule is on.)
 - If you did not block SSH as your first rule, enter `sudo ufw status` and determine the line number of your `deny 22` rule. Add that line number to the end of the command.
 - You will be prompted to delete your `deny 22` rule.

Bonus:

10. You have blocked all incoming traffic except for the protocols deemed essential. Try to SSH once more from the firewalld container into UFW. What happens? (We'll learn more about firewalld in the next section.)

- You'll notice you're still blocked even though we deleted the SSH rule, this is because we are still blocking all incoming traffic. We will now add a rule to just allow only the firewalld container to SSH into UFW.
- Now you will allow the firewalld's IP address to access your container via SSH.
 - Type the command that allows traffic to `ufw container` on port `22` from firewalld containers IP address.

To recap, you now have a rule in place that blocks all incoming traffic and a rule allowing firewalld container's IP address to connect on port `22` to UFW container.

11. Switch back to the firewalld container and test using SSH again:

- From the firewalld container, try to SSH back into the UFW container.
- Type the command to SSH into `UFW container`.
- Exit the active SSH connection.